Alisya Sofiyuddin

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Education

Imperial College London

MEng *Mechanical Engineering with Nuclear Engineering* *September 2023 – June 2027*

- First Class Honours Expected (Highest UK Classification)
- Relevant Modules: Stress Analysis, Fluid Mechanics, Thermodynamics, Heat Transfer, Machine Dynamics and Vibrations, Mechatronics, Computing, Nuclear Thermal Hydraulics
- Final Year Project: Thermomechanical Analysis of Hypersonic Propulsion Systems.

Experience

MuWave Cooling Loop Design Intern | UK *July 2026 – August 2026*

UK Atomic Energy Authority (UKAEA) FOSTER Program

- Developed a conceptual water-cooling system for a future gyrotron test facility, defining thermal requirements, evaluating cooling strategies and selecting a hybrid buffer-tank and air-cooled chiller configuration.
- Developed a Python thermal-hydraulics tool for component sizing, pressure-drop analysis and finite-volume modeling of coolant temperature.
- Performed sensitivity analysis on key design parameters and developed the Process Flow Diagram and preliminary system layout.

Amazon Web Services Data Center Engineering Intern | UK *July 2025 – September 2025*

- Developed an interactive HTML/CSS internal web tool to improve engineer access to complex mechanical and electrical system information.
- Supported preventive maintenance and root-cause analysis of chillers, cooling towers, pumps, generators, UPS/PDUs and switchgear, and analyzed HVAC, BMS and EPMS data to assess cooling demand and reliability.

Malaysia Airlines Engineering and Maintenance Intern | Malaysia *July 2022*

- Interpreted aircraft maintenance documentation and assisted with Airbus A330 maintenance activities involving propulsion, hydraulic and electrical systems, including Rolls-Royce Trent fan blade removal procedures.

Projects

Volumetric Additive Manufacturing 3D Printer | Arup Prize *October 2025 – June 2026*

- Led a multidisciplinary engineering team and designed an optomechanical projection system using Ansys OpticStudio, integrating optical simulation, mechanical design and validation.
- Developed a Python image-predistortion algorithm, reducing experimentally measured barrel distortion by approximately 60%.

Pressurized Water Reactor (PWR) Thermal-Hydraulics Model *September 2025*

- Developed a Python thermal-hydraulics model simulating steady-state, single-phase coolant flow, convective heat transfer and pressure drop in a PWR subchannel.
- Calculated axial coolant and cladding surface temperature distributions using turbulent-flow correlations and a cosine axial power profile; developed a Tkinter GUI for parameter input and results visualization.

Gibbon Conservation Project *September 2024 – June 2025*

- Designed and prototyped a crane for gibbon habitat connectivity, performing structural analysis using Abaqus and SOLIDWORKS and comparing FEA results against analytical calculations to assess the design.
- Manufactured structural components using milling, turning, CNC machining, spot welding and 3D printing; contributed to assembly and experimental testing of the complete prototype.

Skills

- **Programming:** Python, MATLAB, Simulink, Git/GitHub, HTML/CSS
- **Engineering Software:** SOLIDWORKS, Abaqus FEA, Ansys OpticStudio, Ansys Granta
- **Languages:** English (Native), Malay (Native), French (Elementary)